

# SUMIT KUMAR

AI / Generative AI Engineer

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## SUMMARY

AI Engineer with 2+ years building production Generative AI systems — multi-agent orchestration, retrieval-augmented generation, and multimodal generation — for enterprise clients including **Mankind Pharma** and **Bharat Serums and Vaccines**. Owns delivery end to end: LLM architecture, evaluation and confidence design, and secure deployment on Microsoft Azure with CI/CD and production monitoring.

## EXPERIENCE

### AI Engineer — ThoughtSol Infotech Ltd., Delhi, India

Feb 2026 – Present

- Built and shipped a generative media studio now used by **244 users across all 16 Mankind Pharma divisions** — unifying Gemini image generation, Veo and Seedance 2.5 video, and ElevenLabs TTS behind a single Azure-hosted Python backend, with ffmpeg pipelines for video assembly and voiceover.
- Engineered a **layout-preserving poster localization pipeline** using Gemini image editing to translate on-image text into Indian regional languages while retaining original design, typography, and composition — replacing manual redesign for every regional campaign variant.
- Developed a computer-vision compliance engine validating health-camp image submissions against 25+ business rules at **~4s per image** — **88% agreement with human reviewers on a 146-document ground-truth set**, cutting manual review effort ~80% — using an ensemble of GPT vision (Azure OpenAI), a custom Azure Vision model, Document Intelligence, and Custom Face API / Gemini identity and anti-tampering checks.
- Designed that engine's **evaluation and confidence layer**, replacing binary pass/fail output with confidence-scored three-state routing (logprobs, self-consistency, deterministic validation) so only low-confidence cases reach human review; validated a production model migration on a **200-event benchmark suite**, with division-level error analysis isolating systematic failure modes in logo detection and hedged language.
- Built a LangGraph voice agent (ElevenLabs STT/TTS) that runs doctor surveys autonomously at **~700ms response latency**, streaming in real time with barge-in and dynamic follow-up questions — removing months of field-rep survey training.
- Built an AI market research pipeline for **Bharat Serums and Vaccines** that extracts structured data from HCP interview transcripts into cumulative Excel reports, handling piped/carry-forward, matrix, and tick-table question structures across multiple survey instruments — delivered as FastAPI microservices on Azure.
- Delivered all systems inside a secure private Azure architecture (VNet, WAF, API Management, Key Vault) with a dev/UAT/prod release strategy, environment-scoped configuration, CI/CD, and Application Insights usage and performance monitoring.

### AI Engineer — India Career Support (Freelance), Remote

Dec 2025 – Feb 2026

- Architected and shipped a production career-support assistant on a **hybrid RAG layer (Qdrant + MongoDB)** with a LangGraph/LangChain multi-agent workflow, delivered as FastAPI microservices on Azure Container Apps and Functions with CI/CD — cutting manual counselor effort ~45% across thousands of monthly sessions.

### AI Engineer — AiRA, Bengaluru, India

Sep 2024 – Nov 2025

- Built and deployed end to end (backend + Azure infrastructure) a conversational AI platform where users chat with distinct AI personalities backed by **persistent per-user memory on a Memgraph knowledge graph** and MongoDB — sustaining coherent context across sessions at **200 peak concurrent users** on OpenAI and Gemini LLMs.
- Designed a LangGraph multi-agent architecture with specialized agents for memory, retrieval, reasoning, persona, summarization, and orchestration; selected **Memgraph over Neo4j for in-memory traversal latency at scale**, keeping memory lookups off the conversational critical path as graph size grew.

### Software Development Trainee — Udyat Technologies, Mohali, India

Mar 2024 – Sep 2024

- Integrated GPT-4 and Claude with backend APIs using RAG and prompt engineering (~30% accuracy gain); built an AI resume-parsing API (Tesseract OCR + NLP) that improved document-processing speed ~40%.

## TECHNICAL SKILLS

**Languages & Databases:** Python, SQL, MySQL, PostgreSQL, MongoDB, NoSQL, Azure Cosmos DB

**Generative AI & Agents:** LLMs, Generative AI, Agentic AI, Multi-Agent Systems, RAG, GraphRAG, Prompt Engineering, LLM Fine-Tuning (LoRA/QLoRA), NLP, Computer Vision, Multimodal AI

**Frameworks & Tools:** LangChain, LangGraph, Model Context Protocol (MCP), Hugging Face, PyTorch, OpenCV, Tesseract OCR, ffmpeg

**Retrieval & Knowledge Graphs:** Qdrant, Azure AI Search, embeddings, vector search, hybrid search, reranking, Memgraph, Neo4j, Knowledge Graphs

**Backend & APIs:** FastAPI, REST APIs, async programming, Pydantic, Microservices Architecture

**Cloud & MLOps:** Microsoft Azure (Azure OpenAI, AI Search, AI Vision / Custom Vision, Document Intelligence, Container Apps, Functions, Blob Storage, API Management, Key Vault), Docker, Azure DevOps, CI/CD, Git

**Models:** OpenAI GPT, Anthropic Claude, Google Gemini, Veo, Seedance 2.5, ElevenLabs

**Evaluation & Monitoring:** Ground-truth benchmarking, accuracy and error analysis, confidence scoring, LLM evaluation, Azure Application Insights

## EDUCATION

B.E., Computer Science (Specialization in AI & Machine Learning) — Chandigarh University

2021 – 2025